Pension Fund Investment Risk

Portfolio Allocation Analysis for a Municipal Retirement Fund

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Executive Summary

This report evaluates portfolio allocation strategies for a closed municipal retirement fund with a 30-year payment horizon. The fund currently holds a 60/40 equity-to-fixed-income split and faces the central question of whether this allocation adequately protects against insolvency while preserving a reasonable ending surplus. We conducted 1,000 market simulations using the Actuaries' Interest Rate Generator and an Economic Scenario Generator, testing every allocation from 0/100 to 100/0 across three risk metrics.

1. Historical Trends

Historical Trends. A 60/40 portfolio has performed well over the past century, though it suffered significant losses during the Great Depression, the 2008 financial crisis, and the COVID-19 pandemic. The 2022 downturn was an unusual case driven by an unprecedented pace of interest rate increases.

2. Risk of Severe Loss.

Simulation and historical data confirm that annual losses exceeding 15% are rare, occurring in roughly 1.5–3.4% of scenarios. These losses are driven by extraordinary macroeconomic events, not routine market cycles.

3. Portfolio Choice

Evaluated across insolvency, volatility, and ending surplus, the 45/55 allocation achieves the lowest insolvency rate at 21.9%, reduces volatility by approximately 50% relative to the current 60/40 split, and produces an expected surplus sufficient for the city's needs.

4. Adding $175 million

Injecting $175 million reduces insolvency risk to nearly 0% across most allocations. With survival effectively guaranteed, a 60/40 split becomes the recommended choice—its higher expected returns compensate the city for the capital outlay.

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I. Historical Record

Before recommending a portfolio allocation, it is important to establish what kind of losses the fund should prepare for. This section examines the rolling 12-month returns of a 60/40 portfolio over nearly 100 years of market data. Rolling returns smooth out month-to-month noise and offer a more realistic picture of what a long-term investor actually experiences in any given year.

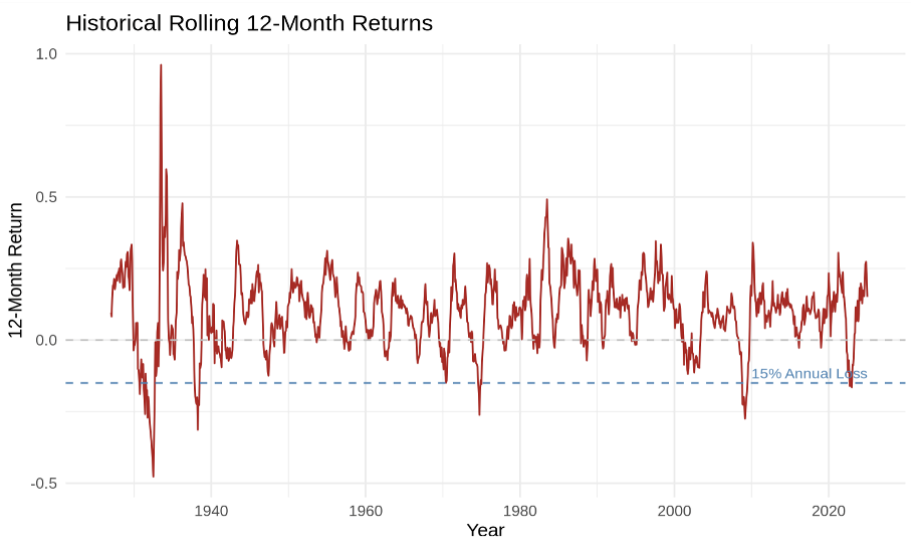

Figure 1 - Rolling 12-Month Returns of a 60/40 Portfolio (1926–2024)

The dashed line at –15% marks a threshold of severe loss. Over this entire period, only five episodes breached that line: twice during the Great Depression and the run-up to World War II, once during the 1970s oil crisis and Vietnam War era, once during the 2008 financial crisis, and once in the early 2020s during the COVID-19 pandemic. Each of these events was global in scale and driven by forces well outside normal market cycles. Outside of these episodes, the 60/40 portfolio has recovered reliably, with most 12-month windows producing positive returns.

The fund's most recent downturn produced the worst 12-month returns in nearly two decades. The cause was unusual: after the Federal Reserve lowered interest rates to near zero during the COVID-19 pandemic, rates were reversed sharply throughout 2022. Rates were rising several percentage points within months. Both equity and bond prices fell simultaneously, making 2022 a uniquely punishing year for balanced portfolios.

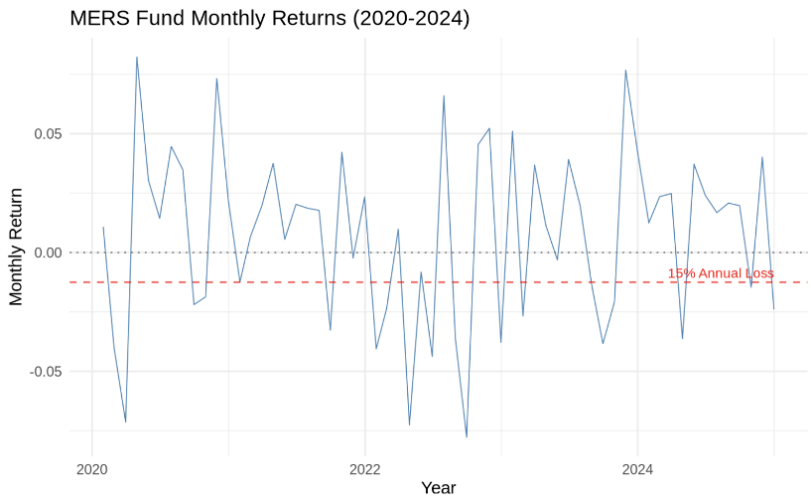

Figure 2 - Fund Monthly Returns (2020 - 2024)

In 2023 and 2024, monthly returns remained volatile, alternating between gains and losses with little consistency. Full year returns for both years were positive, but the path to those returns was uneven; a pattern consistent with an economy still adjusting to post-pandemic conditions. The red dashed line in Figure 2 represents the monthly equivalent of a 15% annual loss, providing context for how individual monthly swings compare to the severe-loss threshold.

II. Probability of Severe Annual Loss

A natural concern for any retirement fund is the likelihood of a catastrophic year-one in which portfolio losses exceed 15% within a 12-month window. This section quantifies that risk using both historical data and simulation of the future.

As shown in Figure 1, losses of this magnitude occurred in only five distinct episodes across nearly 100 years of data. Each was triggered by an extraordinary macroeconomic event: a global depression, a geopolitical crisis, or a pandemic. These are events that affect entire economies, not the result of poor investment decisions.

To complement the historical record, we ran 1,000 simulations of the 60/40 portfolio using an Economic Scenario Generator. The table below compares the probability of a 15%+ annual loss under both methods.

Table 3: 15%+ Losses in Return

Method	Probability
Scenario	1.51%
Historical	3.4%

Table 1 – Probability Losses Exceed 15% in 12 Month Period

The simulated probability (1.51%) is lower than the historical figure (3.40%), which is expected: the ESG models a broader distribution of outcomes, diluting the weight of the historical tail events. Both figures confirm that losses of this severity are marginal, occurring fewer than once every 30 years, on average. The primary threat to the fund is not a single catastrophic year, it is the cumulative effect of underperformance over the 30-year period. The insolvency and volatility metrics in the following section capture this risk directly.

III. Portfolio Recommendation

The fund's obligation is to make 30 years of scheduled payments to retirees without exhausting its balance. Any surplus remaining after those payments can be directed toward other city needs, but generating a large surplus is secondary to meeting the fund's core commitment. With this priority in mind, we evaluated every allocation from 0/100 (all fixed income) to 100/0 (all equity) across three metrics: insolvency risk, volatility of the ending balance, and the distribution of the ending surplus.

All results below are based on 1,000 simulations per allocation using the Actuaries' Interest Rate Generator and an Economic Scenario Generator in Excel. The model accounts for adverse, neutral, and favorable market conditions across the full 30-year period.

Insolvency

Insolvency is the most consequential metric for this fund. It measures the share of simulations in which the fund's balance reaches zero before all 30 years of payments have been made. A fund that becomes insolvent cannot meet its obligations to retirees; it’s a scenario with serious legal and social consequences for the city.

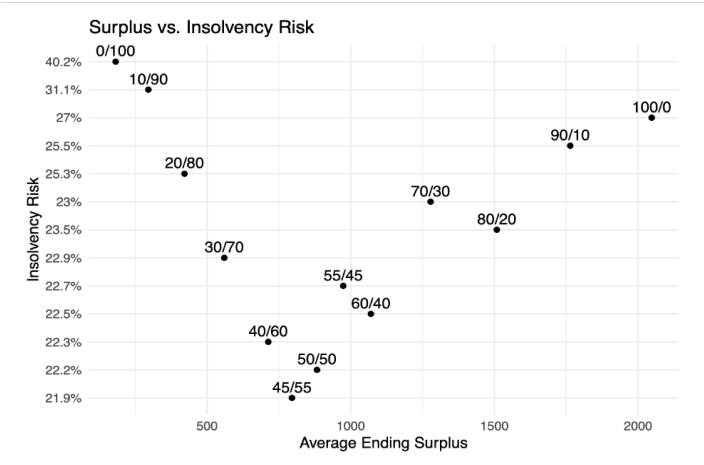

Figure 4 – Ending Balance vs Insolvency Tradeoff

Figure 4 reveals a characteristic V-shaped relationship between allocation and insolvency. This shape has an important implication: both extremes of the allocation spectrum carry elevated risk, though for different reasons.

At the conservative extreme (0/100, 10/90), insolvency rates are highest, reaching 40.2% for an all-fixed-income portfolio. This may seem counterintuitive, since bonds are typically considered safe. The explanation is that a portfolio composed entirely of fixed-income assets does not

generate sufficient returns to keep pace with 30 years of scheduled payouts. The fund slowly bleeds capital and in a significant share of scenarios, it runs dry before the obligation period ends.

At the aggressive extreme (90/10, 100/0), insolvency rises again to 25–27%. Here the risk is different: equity-heavy portfolios have the potential for high returns, but they are also vulnerable to sharp drawdowns. A few consecutive bad years early in the fund's life can deplete the balance beyond recovery, even if markets eventually rebound.

In the middle range (40/60 through 55/45), insolvency reaches its minimum. The 45/55 allocation sits at the bottom of the curve at 21.9%, with the 50/50 and 40/60 allocations close behind. These portfolios balance sufficient return generation against manageable downside exposure. The current 60/40 split performs reasonably at 22.5%, but the additional 15 percentage points of equity exposure adds risk without a proportionate reduction in insolvency.

Recommendation: shift from 60/40 to 45/55. This allocation achieves the lowest insolvency rate in the simulation set while maintaining an average ending surplus near $800 million. That is more than sufficient for the fund's secondary purpose of supporting city operations.

Volatility

Insolvency measures whether the fund survives; volatility measures how widely the outcomes vary across the 1,000 simulated scenarios. Each simulation reflects a different path of market returns, some favorable and some adverse. As a result, the ending balance is not a single number but a distribution of possible outcomes. Volatility captures the spread of that distribution.

A portfolio with high volatility can produce ending balances that range from near-zero to several billion dollars, even if its insolvency rate is low. This kind of spread makes it difficult for the city to plan budgets or allocate resources around the fund's expected trajectory. A narrower distribution, or lower volatility, gives the city more confidence that the fund's actual balance will land close to its projected value.

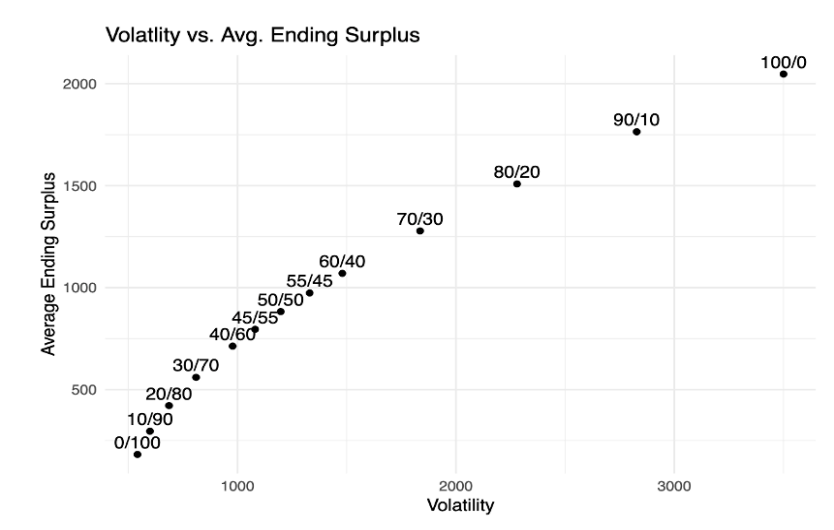

Figure 5 - Average Ending Surplus vs. Volatility by Allocation

Figure 5 plots each allocation's average ending surplus against the standard deviation of that surplus across all 1,000 simulations. The relationship is approximately linear through the moderate allocations and then accelerates sharply for equity-heavy portfolios. Moving from 45/55 to 60/40 increases the average surplus by 30%, from roughly $850 million to $1.1 billion. However, the standard deviation gets nearly a 40% increase from approximately $1.1 to $1.5 billion. Moving further to 80/20 or 90/10 produces even larger surpluses on paper, but with standard deviations exceeding $2.5 billion, the range of possible outcomes spans from deep bankruptcy to windfalls of several billion dollars.

For the fund's purposes, a high-volatility portfolio is problematic even if its average outcome is favorable. The 45/55 allocation reduces volatility by approximately 40% relative to the current 60/40 split. This means the range of likely outcomes narrows considerably, making the fund's trajectory far more predictable and its obligations far more secure.

Ending Surplus Distribution

The previous two metrics summarize performance with single numbers: a rate and a standard deviation. The histogram below provides a fuller picture by showing how the 1,000 simulated ending balances are distributed under the recommended 45/55 allocation.

The shape of the distribution is clearly right-skewed. The average ending balance is around 800 but this has been increased by the outliers. Most simulations fall between 0-400; the average doesn't represent the most probable outcome.

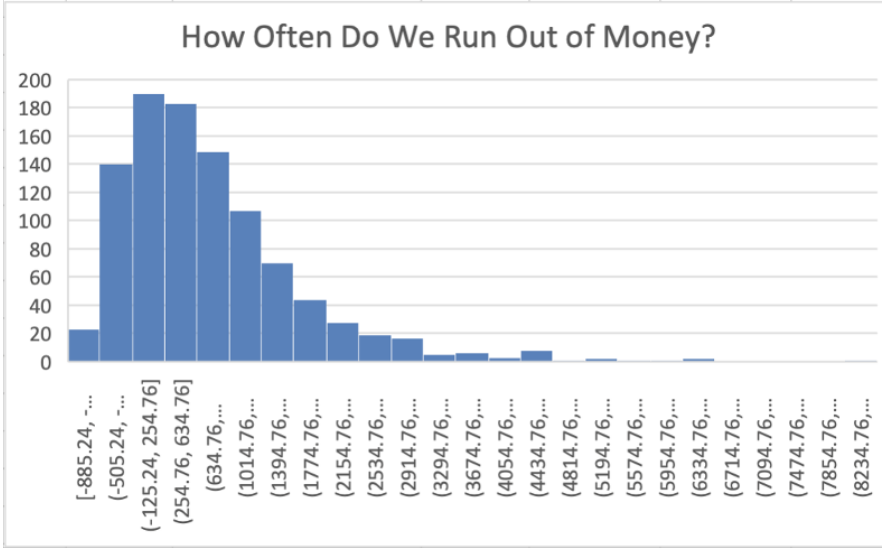

Figure 6 - Distribution of Ending Fund Balance After 30 Years (45/55 Allocation)

The distribution is heavily right-skewed. The average ending balance is approximately $800 million, but this figure is misleading: it is pulled upward by a long tail of high-surplus scenarios that occur infrequently. The majority of simulations cluster between $0 and $400 million, and the most common outcome falls in the $100-$250 million range. An ending balance in the $150-$300 million range is more than adequate for the fund's secondary objective of supporting city operations, and the 45/55 allocation delivers this range with the highest reliability.

IV. Portfolio Recommendation with $175M Top-Up

This section considers an alternative scenario in which the city injects an additional $175 million into the fund at the outset. This contribution represents a significant short-term strain on city finances, but it fundamentally changes the fund's risk profile and opens the door to a higher-return allocation.

Insolvency After Top-Up

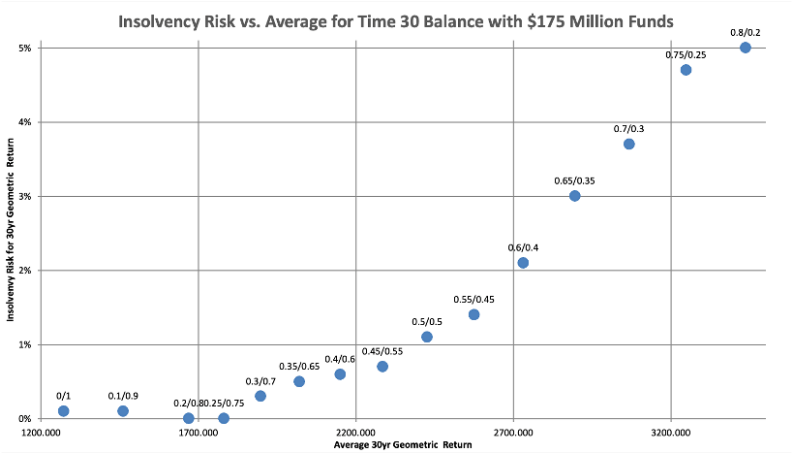

Figure 7 - Insolvency Risk and Ending Balance by Allocation After $175 Million Injection

The impact on insolvency is dramatic. With the additional capital, allocations from 0/100 through 45/55 drop to near-zero insolvency rates. Even the 60/40 portfolio, which carried a 22.5% insolvency rate without the top-up, falls to approximately 2%. The additional $175 million acts as a buffer that absorbs early-period losses, giving the fund time to recover from adverse market conditions that would otherwise deplete it.

This shift has an important strategic implication. Without the top-up, the fund's primary concern is survival, and the optimal allocation is conservative (45/55). With the top-up, survival is effectively guaranteed across all reasonable allocations, and the decision shifts to a different question: how much surplus should the fund aim to generate?

Volatility After Top-Up

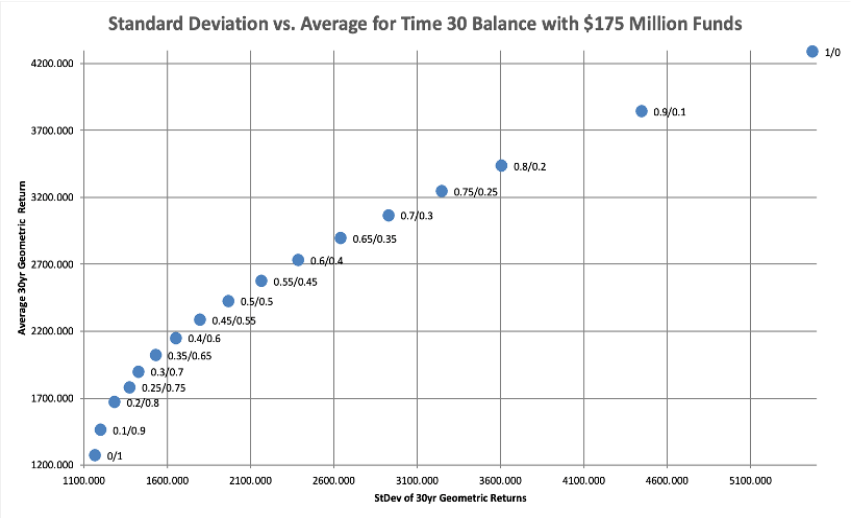

Figure 8 - Standard Deviation vs. Average Ending Balance After $175 Million Injection

Figure 8 shows the familiar return-volatility tradeoff, but with a crucial difference: the entire curve has shifted upward. Even conservative allocations now produce meaningful surpluses, and the moderate allocations deliver ending balances well above what the fund needs. The 60/40 portfolio, for example, projects an average ending balance near $2.8 billion with the standard deviation at $2.6 billion. While that standard deviation is large, one standard deviation below the mean still sits above zero. The fund can tolerate this level of volatility because the $175 million buffer protects against the worst outcomes.

Recommendation with the top-up: shift to 60/40. With insolvency risk effectively eliminated, the fund can afford to pursue higher returns. The 60/40 allocation's projected surplus compensates the city for the $175 million outlay and generates additional capital for municipal use. The city would face a short-term budget constraint, but the fund's improved trajectory provides long-term financial flexibility that more than offsets the initial cost.

V. Conclusion

Without additional funding, the 45/55 allocation is the strongest choice. It achieves the lowest insolvency rate (21.9%), reduces volatility by half relative to the current allocation, and produces an ending surplus distribution centered in a range that meets the fund's needs.

With a $175 million injection, the 60/40 allocation becomes optimal. The additional capital eliminates meaningful insolvency risk, and the higher expected returns justify the city's short-term sacrifice. The fund transitions from a survival-focused posture to one that can reliably generate surplus for broader municipal benefit.

In either scenario, the fund's first obligation, meeting 30 years of promised payments to retirees, is well protected. The choice between the two paths depends on whether the city is willing and able to commit the additional capital.